Speaker Script

Bimodal Affective Alignment

Enhancing Conversational Empathy through Textual and Facial Fusion

Course	HCI / Affective Computing — Course Project Presentation
Authors	Sainadth Pagadala · Vishakan Umapathy · Dr. Minhua Huang (Advisor)
Institution	College of Engineering and Computer Science Texas A&M University–Corpus Christi
Total slides	20 · Approx. 18–22 minute talk

Slides 1 – 10	**Vishakan Umapathy / Sainadth Pagadala** Title, problem framing, contributions, pipeline, formulation, label mapping, models, demo, headline benchmark, fine-tune.
Slides 11 – 20	**Co-presenter** Dissonance, α-sensitivity, latency, qualitative results, replies, fine-tuned downstream wins, failure modes, limitations, takeaways, conclusion.

HOW TO READ THIS SCRIPT

Each slide entry has three parts. **ON SCREEN** reminds you what the audience is currently looking at, so you don't have to glance back at the projector. **WHAT TO SAY** is the actual narration — written for the spoken voice, not written prose, so it should sound natural read aloud at a calm pace. **TRANSITION** is the line that bridges into the next slide; it lets you click forward without an awkward pause.

Pacing target: roughly one minute per slide on average. Some slides (10, 16, 19) deserve a beat longer; some (3, 6, 18) can be moved through quickly. A few sentences are *italicised*; those are optional asides you can drop if you're running short on time.

01
of 20

Title

Authors, affiliation, the model stack

ON SCREEN

Dark navy title slide. Project name in large serif type, subtitle below it, author list on the left, university on the right, and a tag line listing RoBERTa, ResNet-50, late fusion, FLAN-T5, and Streamlit.

WHAT TO SAY

Good [morning / afternoon] everyone. I'm Vishakan, and together with Sainadth — and under the guidance of Dr. Minhua Huang — we built a system called Bimodal Affective Alignment.

The full title is *Enhancing Conversational Empathy through Textual and Facial Fusion*, and that subtitle actually does most of the work for us today. We're going to show you a dialogue system that listens to two channels at once — the words a user types, and what their face is doing in the same moment — and we're going to show you what changes when the system can read both signals instead of just one.

The tag line at the bottom previews the stack: a RoBERTa text classifier, a ResNet-50 face classifier, a single-parameter fusion rule, FLAN-T5 generating the reply, and a Streamlit interface that exposes every intermediate signal so a reviewer can watch the system reason.

TRANSITION

Let me start with the problem we set out to solve.

02
of 20

The Problem

A unimodal system misreads the room

ON SCREEN

Two side-by-side cards: 'Text alone is easy to fool' and 'Vision alone has the mirror weakness'. Below, a navy banner with our claim in bold: reading both channels gives a second opinion, and a way to flag disagreement.

WHAT TO SAY

A dialogue system that only sees text is easy to fool. Sarcasm reads as cheerful. A polite "I'm fine" hides distress. A deadpan one-liner gives almost no signal at all. And in every one of those cases, the system has to guess — and a wrong guess in an empathy loop costs user trust very quickly.

Vision on its own has the mirror weakness. The same neutral face shows up in good lighting, in bad lighting, off-axis, partially occluded. So a face classifier in isolation also has to guess.

Our claim is in the navy bar at the bottom. Reading both channels together does two things. First, when both branches agree, you get a second opinion — your decision is more reliable. Second, and this is the part we care about for HCI, when the two channels disagree, you have a principled way to *flag* that disagreement instead of silently picking one and hoping.

TRANSITION

So that's the motivation. Here's what we actually built and contributed.

03

of 20

Approach & Contributions

Five concrete deliverables

ON SCREEN

Five numbered items, each with a coloured circle: the bimodal pipeline, label-space alignment, end-to-end Streamlit demo, FER2013 fine-tune, and reproducible benchmarks.

WHAT TO SAY

Five concrete deliverables. The first is the pipeline itself — text plus face, into a seven-way affect decision, into an empathetic reply.

The second is label-space alignment. The text dataset has 28 emotion labels; the face classifier has 7. We had to make those agree before fusion was even meaningful, and we did it with a deterministic mapping I'll show you in a couple of slides.

Third, an end-to-end Streamlit demo that exposes every intermediate signal. We could have hidden the probability vectors and just shown the final reply — but the whole point is that a reviewer should be able to *watch* the system reason, not just trust the output.

Fourth — and this is our strongest engineering result — we fine-tuned the face branch on FER2013 and lifted top-1 accuracy from 23.1 percent to 68.5 percent.

And fifth, every number you'll see today comes out of a reproducible script in our repository. Ablation, alpha sensitivity, dissonance, latency — all scripted, all in the harness.

TRANSITION

Let me walk you through the pipeline itself.

04

of 20

System Pipeline

Figure 1 from the report — both branches align before fusion

ON SCREEN

Block diagram. Two inputs on the left — utterance U and image frame I — feeding into RoBERTa and ResNet-50, then into orange and teal probability vectors, into a navy 'late fusion' box that emits b, and finally down into a FLAN-T5 box that produces the empathetic reply R.*

WHAT TO SAY

This is Figure 1 of the report. Two inputs come in on the left: U is the user's utterance, and I is the image frame from a webcam or upload.

Each input goes to its own sensor. U goes to a RoBERTa transformer fine-tuned on GoEmotions; the image goes to a ResNet-50 with a seven-way head. Both branches emit a probability vector over the same seven FER classes — that's the orange P-text and the teal P-face.

Those two vectors go into the navy fusion block, which combines them using a single trust weight alpha, and produces a discrete decision b-star — that's the fused emotion class.

Finally b-star and the original utterance U get passed to FLAN-T5, which writes the empathetic reply R. One thing worth flagging: the probability vectors themselves are *never* exposed to the language model. Only U and the class string for b-star feed the prompt. We did that on purpose, to keep the response stage clean.

TRANSITION

Now let's look more closely at that fusion block — it's just one parameter.

05
of 20

Late Fusion: A Single Trust Knob α

Equation 1 — one parameter, three operating points

ON SCREEN

The fusion equation in a navy banner: b equals argmax over k of alpha times P-text plus one minus alpha times P-face. Below it, three cards for the three operating points: alpha equals 1, 0.5, and 0.*

WHAT TO SAY

The fusion rule is Equation 1, and it's deliberately simple. b-star — the fused decision — is the class index k that maximises a weighted average of the two probability vectors.

Alpha is a single number between zero and one. Three settings are operationally meaningful, and they're the three cards across the bottom.

When alpha equals 1, the rule recovers text-only — we ignore the face entirely. When alpha equals 0, we recover face-only — we ignore the utterance. And when alpha equals 0.5, both branches contribute equally. That's the system default, and it's what we treat as the balanced setting.

The line at the bottom is important: a sample is flagged as showing affective dissonance whenever the argmax of P-text doesn't match the argmax of P-face. That's the dissonance flag the Streamlit interface surfaces as a coloured badge — and we'll see real examples of it later.

TRANSITION

Before fusion can work, the two label spaces have to agree. Here's how we made that happen.

06
of 20

Aligning the Label Spaces

28 GoEmotions labels → 7 FER classes

ON SCREEN

Left side: a short explanation of the challenge and our solution. Right side: a table showing how GoEmotions labels collapse into the seven FER classes — for example, anger, annoyance, and disapproval all map to 'Angry'.

WHAT TO SAY

GoEmotions ships with 28 fine-grained emotion strings — things like annoyance, disapproval, amusement, realization. The visual branch only resolves seven. So we had a label-space mismatch, and we couldn't fuse anything until we fixed that.

Our solution is on the right. We hand-coded a surjection — a Python dictionary — that maps every one of the 28 GoEmotions labels onto exactly one of the seven FER classes. So anger, annoyance, and disapproval all roll up to Angry. Joy, amusement, excitement, and love all roll up to Happy. And so on.

One subtle point worth mentioning: GoEmotions is multi-label and uses sigmoid scores, not softmax. So we don't just pick one class — we sum the sigmoid mass landing in each FER bin and renormalize. That preserves the relative confidence of fine-grained labels when several map to the same bin — which is exactly what you want.

TRANSITION

So that's the label space. Let me introduce the three actual models.

07

of 20

Three Models, One Pipeline

Off-the-shelf checkpoints + a deterministic prompt

ON SCREEN

Three coloured cards: orange RoBERTa (text sensor), teal ResNet-50 (face sensor), navy FLAN-T5 base (response generator). Each card lists a short description, the specific HuggingFace checkpoint or decoding parameters, and the output.

WHAT TO SAY

Three models. Each card here lists a short description, the specific public checkpoint we use, and the output of that branch.

On the left, RoBERTa-base, fine-tuned on GoEmotions — this is the SamLowe checkpoint on HuggingFace. We chose it specifically because the assignment asked for BERT-or-RoBERTa on GoEmotions, and using a publicly available fine-tune saved us from training a 28-way head from scratch.

In the middle, the ResNet-50 face branch — Elena Ryumina's face emotion recognition checkpoint, which is convenient because it works out of the box on near-frontal RGB faces. I'll come back to that point in two slides because it's also where most of our error budget lives.

On the right, FLAN-T5 base — 220 million parameters, deterministic beam search with four beams, no-repeat 3-grams, between 18 and 128 new tokens, no sampling. Deterministic on purpose — we want every reply to be reproducible.

The line at the bottom is the design rule we already mentioned: probability vectors are never exposed to the language model. Only U and the fused class string feed the prompt.

TRANSITION

And here's how all of this is wrapped up for an actual user.

08

of 20

Streamlit Interface

Watch the system reason — don't just trust the output

ON SCREEN

Mock interface on the left: a top bar reading 'Bimodal Affect Demo', an alpha slider at 0.50, three side-by-side bar charts (text, face, fused), an 'AGREEMENT' badge, and a generated reply preview. On the right, a bullet list of the features a reviewer can see.

WHAT TO SAY

This is the Streamlit interface. The mock on the left shows what a reviewer actually sees.

At the top, an alpha slider — they can drag it from 0 to 1 in real time. Below it, three bar charts side by side: the text branch's probability vector in orange, the face branch's in teal, and the fused vector in navy. Underneath that, a coloured badge — green when the two branches agree, orange when they're in dissonance — and the generated empathetic reply.

The list on the right is the feature set: probability vectors, fused decision, alpha slider, the dissonance flag, per-stage timings, the reply, and the option to use either a webcam frame or an uploaded image.

We could have hidden a lot of this and just shown the final reply, but the whole point of the interface is that a reviewer should be able to *probe* the system. Slide alpha, see what changes. Watch where text and face disagree. The system isn't a black box.

TRANSITION

So that's the system. Now — does it actually work? Let's start with the headline numbers.

09

of 20

Quantitative Benchmark · FER-7 Top-1

Table 1 from the report — baseline ResNet-50 face branch, alpha = 0.5

ON SCREEN

Left: a five-row results table. Text-only is 69.7 percent. Image-only is 23.1 percent (in coral). Text plus image fused is 81.6 percent (in teal, bold). The two ablation rows show what each branch says on the same fused pairs. Right: three takeaway callouts.

WHAT TO SAY

This is Table 1 — the quantitative benchmark. Three things to look at.

First, the text branch is solid. 69.7 percent top-1 on the full GoEmotions test split, mapped into the seven FER classes. That's a believable, usable signal.

Second — and this is the bottleneck we'll spend a lot of time on later — the face branch on its own only gets 23.1 percent. That's near chance for seven classes. The reason is domain shift: the AffectNet-trained checkpoint is being asked to score 48 by 48 grayscale FER2013 frames upscaled to 224 by 224. That's not what it was trained on.

Third, the fused row — 81.6 percent. Notice that's just one point below the text-only row of 82.6 on the same image-plus-template pairs. So at the baseline visual checkpoint, fusion is essentially as good as text alone, with the added benefit of being *face-aware* — which is exactly what the dissonance flag needs.

TRANSITION

The face branch was clearly the weak link. So we did something about it.

10

of 20

Fine-tuning the Visual Branch

Table 2 — closing the AffectNet → FER2013 domain gap

ON SCREEN

Three big stat callouts: BASELINE 23.1% in muted gray, an arrow, then FINE-TUNED 68.5% on navy, and a coral badge reading +45.4 percentage points, approximately 3-times improvement. Below, a soft callout explaining the domain shift.

WHAT TO SAY

This is the strongest engineering result in the report, and I want to spend a moment on it.

The default visual checkpoint was fine for a live demo, but it was clearly the weakest part of the pipeline on FER2013 test data. So we fine-tuned the same ResNet-50 head on the FER2013 train split with a held-out validation slice and standard cross-entropy.

Top-1 accuracy moved from 23.1 percent to 68.5 percent. That's plus 45 percentage points, roughly a three-fold improvement, and roughly the difference between near-chance and a usable face signal.

The bottom band explains why. FER2013 test images are 48-by-48 grayscale faces upscaled to 224-by-224. That's a noticeable domain shift from the AffectNet distribution that the off-the-shelf checkpoint was trained on. Fine-tuning closes most of that gap.

One important note: we deliberately keep the fine-tuned weights as a *separate* checkpoint from the deployed demo. So every baseline number you've seen so far stays reproducible without extra setup, and we'll show in a few slides what changes downstream when we swap the fine-tuned checkpoint in.

TRANSITION

And with that, I'm going to hand things over to my co-presenter, who'll take you through what happens when text and face disagree, the timing budget, and the qualitative results.

HANDOFF — END OF FIRST PRESENTER

Slides 11 through 20 are presented by the second co-presenter.

Suggested handoff line (already at the end of slide 10's script):

> *"And with that, I'm going to hand things over to my co-presenter, who'll take you through what happens when text and face disagree, the timing budget, and the qualitative results."*

Suggested opening line for the second presenter on slide 11:

> *"Thanks. So far we've shown what happens when the two channels are aligned. Let me show you what happens when they aren't."*

11

of 20

Dissonance Evaluation

Table 3 — what happens when text and face disagree?

ON SCREEN

A two-row results table. The 'real FER2013 + mismatched text' row shows alpha-1 at 4.0%, alpha-0.5 at 4.3% (highlighted in coral), and alpha-0 at 23.1%. The synthetic row shows alpha-0.5 and alpha-0 both at 100% (in teal). Below, two insight cards.

WHAT TO SAY

Thanks. So far we've shown what happens when the two channels are aligned. Let me show you what happens when they aren't.

We ran two complementary probes. In both cases, we deliberately mismatch the text with the gold face class — the formula at the top describes how — so the system is being given a face and a contradicting sentence, on purpose.

The first row is the deployment-relevant one: real FER2013 images paired with deliberately mismatched template sentences, all 3,589 test rows. At alpha equals one, the system follows the misleading text 96 percent of the time, which is exactly what you'd expect — alpha equals one means trust the text. At alpha equals 0.5 the fused decision still mostly follows text. Why? Because the baseline face branch is only 23.1 percent accurate, so it's not a reliable enough signal to outweigh a confident, wrong text branch. That's a real limitation of the baseline setup.

The second row is the sanity check. We replaced the face branch with a perfect one-hot vector for the intended class — so one input is wrong by construction, and the other is perfect. At alpha equals 0.5, the fused rule recovers the intended class 100 percent of the time. That confirms the fusion *arithmetic* works exactly the way it's supposed to. The bottleneck in row one is the noisy face branch, not the rule.

TRANSITION

That brings up the obvious next question — should we just tune alpha higher or lower? Let me explain why we don't.

12

of 20

Why α is an HCI Knob, Not a Hyperparameter

Interpretable trust setting — exposed on the sidebar

ON SCREEN

Bar chart on the left showing the matched-bimodal benchmark at the three operating points: alpha-1 at 82.6 percent, alpha-0.5 at 81.6, alpha-0 at 23.1. On the right, a short explanation of why we expose alpha rather than tuning it.

WHAT TO SAY

Equation 1 is a one-parameter family of decision rules, and three points along it are operationally meaningful. At alpha equals one, text-only, we get 82.6 percent. At alpha equals 0.5, fused, 81.6. At alpha equals zero, face-only, 23.1.

We could have tried to find the optimal alpha on a held-out set — but that's not actually what alpha is for in this system.

Alpha is meant to be an interpretable HCI knob, not a hyperparameter to tune. It's exposed on the Streamlit sidebar so a reviewer can probe the same input under different trust regimes — slide alpha up to trust the text, slide it down to trust the face — and watch the dissonance badge change in real time.

Alpha equals 0.5 is a sensible default because it inherits whichever branch is more confident. Remember the synthetic dissonance result on the previous slide — at 0.5, with a perfect face, we recovered the intended class 100 percent of the time, even though the text was wrong by construction. So 0.5 isn't a compromise — it's a deliberately balanced position.

TRANSITION

We talked about quality. Let me also show you that the system is actually fast enough to use.

13
of 20

Latency Benchmark

Table 5 — Apple MPS, 15 timed runs after 3 warm-ups

ON SCREEN

A timing table on the left: text model 18.9 ms, face model 19.4 ms, fusion essentially zero, affect subtotal 38.4 ms (in teal), FLAN-T5 generation 904 ms, end-to-end 942 ms. On the right, a teal callout showing 38 ms affect inference vs. the 400 ms Doherty threshold, and a card explaining that generation dominates.

WHAT TO SAY

This is the latency breakdown — Table 5 in the report. We benchmarked on Apple MPS with 15 timed runs after 3 warm-ups.

The text model takes about 19 milliseconds. The face model takes about 19 milliseconds. The fusion arithmetic itself is essentially free — 0.1 milliseconds.

So affect inference — the part where the system decides which emotion class — finishes in about 38 milliseconds. The teal callout puts that in context: that's comfortably under the 400-millisecond Doherty interaction threshold from the HCI literature. So a user holding the alpha slider sees the bar charts update with no perceptible lag. That part of the system is fast.

Generation, however, dominates. FLAN-T5 takes about 900 milliseconds, which puts the end-to-end at about 942. That's because we're using deterministic beam search with four beams on a 220-million-parameter model on a laptop GPU.

The card on the right notes the production fix: in a deployed product, the response would be *streamed* token-by-token, so the user would perceive a sub-400 millisecond onset even though the full message lands later. We didn't implement that — but it's a known design move.

TRANSITION

Now let me show you what the system actually does on real images, class by class.

14
of 20

Per-Class Qualitative Cases

Table 6 — one real FER2013 image per gold class, baseline ResNet-50

ON SCREEN

A table with seven rows — one per FER-7 class — showing the predicted class at alpha equals 1, 0.5, and 0. Six of the seven gold classes are correct at alpha-1 and alpha-0.5; the alpha-0 column collapses to Happy on five of seven. On the right, three takeaway notes.

WHAT TO SAY

This is Table 6. We took one real FER2013 test image per FER-7 gold class, paired it with a matching template sentence, and read out the predicted class at the three operating points.

Two patterns jump out, and they're listed on the right.

First, the face branch is unreliable on FER2013. Look at the alpha-0 column — five of seven rows predict Happy. That's the per-row picture behind the 23.1 percent aggregate we saw earlier. Domain shift, all the way down.

Second, the text branch rescues the fused decision on six of seven rows. The fused column agrees with gold for Angry, Disgust, Fear, Happy, Sad, and Surprise.

The single failure is the Neutral row. The template text is — and I'm quoting — "It is a normal quiet day; I feel **pretty neutral** and nothing in particular is happening to me." The GoEmotions head reads the phrase "feel pretty" as Happy and pulls the whole thing toward Happy. That's a real failure mode of single-utterance sentiment classifiers, and we'll come back to it on the failure-modes slide.

TRANSITION

And here's what the language model actually says in response to those classes.

15
of 20

FLAN-T5 Empathetic Replies

Table 7 — deterministic beam search; only U and b feed the prompt*

ON SCREEN

A seven-row table showing each gold class, the fused class, and the actual generated reply. Five replies are clean acknowledgement plus follow-up question. The Sad row drifts into multiple-choice format and the Surprise row paraphrases the user — both highlighted in coral. A footer summarizes the success rate.

WHAT TO SAY

Table 7. These are the actual FLAN-T5 replies for each of the seven rows we just looked at. Decoding is deterministic, so these are exactly reproducible.

Five of the seven rows are clean. Take the Angry row: *"I hear that you are angry. What do you mean by 'I will not stand for it'?"* That's exactly what the prompt asked for — second-person acknowledgement, validation, and one short open question.

The Fear row, the Happy row, the Disgust row, the Neutral row — all do something similar.

Two failures, both highlighted in coral. The Sad row drifts into a multiple-choice format — *"(i) Yes. (ii) No."* — which is a known habit of FLAN-T5 base on questions it can't quite ground. And the Surprise row just paraphrases the user instead of asking a real question.

We read these as limitations of the 220-million-parameter backbone, not of the alignment pipeline. Swapping in any 1-billion-class instruction-tuned model would be a one-line change in our response synthesizer file, and would cleanly improve generation quality without affecting any of the affect-alignment numbers we just walked through.

TRANSITION

So that's the baseline picture. Now let me show you what changes when we plug in the fine-tuned face branch.

16
of 20

Fine-tuned Face Branch · Downstream Wins

Tables 9 & 10 — from text-dominant to genuinely bimodal

ON SCREEN

Two tables stacked vertically. The matched bimodal benchmark shows fused accuracy moving from 81.6 percent baseline to 87.8 percent fine-tuned. The dissonance benchmark shows fused alpha-0.5 jumping from 4.3 percent to 41.8 percent. A navy banner at the bottom summarizes the headline number.

WHAT TO SAY

Two tables here, both showing what changes when we swap in the fine-tuned face branch. Nothing else changes — same text branch, same fusion rule, same templates.

The top table is the matched bimodal benchmark — Table 9 in the report. Image-only goes from 23.1 percent to 68.5, which we already saw on the fine-tune slide. But notice what happens to the fused row: 81.6 climbs to 87.8 percent. That's the first regime where adding the face actually *helps* the fused decision rather than gently dragging it down. Fused is now clearly above the 82.6 percent text-only ceiling.

The second table is the real-image dissonance benchmark — Table 10. With the baseline face, fused alpha-0.5 was 4.3 percent. With the fine-tuned face, it jumps to 41.8 percent. So on roughly 40 percent of cases where the text is wrong by construction, the fused decision now overrides the text and recovers the intended class. With the baseline branch, that number was essentially zero.

The takeaway in the navy bar at the bottom: fused alpha-0.5 jumps from 81.6 to 87.8 in the matched setting, and from 4.3 to 41.8 under planted-wrong text. That's the visual fine-tune flipping the system from text-dominant into genuinely bimodal.

TRANSITION

With all that said, the system isn't perfect, and we want to be honest about where it still breaks.

17
of 20

Failure Modes (Worth Naming)

Section 5.8 — two qualitative failures show up in our data

ON SCREEN

Two side-by-side cards. Teal card on the left: visual-branch domain shift, with the fix — the FER2013 fine-tune that takes us from 23.1 to 68.5 percent. Coral card on the right: text-branch sentiment leak, with the 'feel pretty' example we saw earlier.

WHAT TO SAY

Two failure modes show up in our data, and they're worth naming because they each motivate a real architectural decision in the system.

The first is visual-branch domain shift — the teal card. The baseline ResNet-50 was trained on color, near-frontal AffectNet faces, and we asked it to score 48-by-48 grayscale FER2013 frames upscaled to 224-by-224. So the model defaults to Happy on five of the seven qualitative rows. That's the 23.1 percent number we kept coming back to. The fix is the FER2013 fine-tune we just covered — that closes most of the gap.

The second is text-branch sentiment leak — the coral card. The GoEmotions head reacts to *surface cue words* rather than to the actual content of the utterance. The Neutral template contains the phrase "feel pretty," and the model classifies the whole sentence as Happy because of that phrase. This is a known weakness of single-utterance sentiment classifiers, and it's exactly the failure mode that the trust-weight alpha is designed to surface — not hide. If a reviewer lowers alpha on this row, they see the face branch (also Happy here, because of the same baseline domain shift) confirm the wrong text decision. The mistake becomes *visible*, not silent. And that's the whole HCI argument for exposing alpha at all.

TRANSITION

A few broader limitations of this study — let me name them up front.

18
of 20

Limitations

Section 5.10 — where this study is incomplete

ON SCREEN

Five numbered cards, in coral: probe not a corpus; scripted dissonance text; visual-branch ceiling; small generator; no formal Likert study.

WHAT TO SAY

Five honest limitations. We want to name them rather than have someone in the audience name them for us.

One — the matched-template benchmark is a probe, not a natural multimodal corpus. It tests whether aligned text and a face estimate can be fused consistently; it doesn't measure how the system handles spontaneous speech.

Two — the dissonance probe is closer to deployment, but it still uses scripted template sentences instead of utterances from real participants.

Three — even after fine-tuning, the visual branch is bounded by FER2013 label noise and by the fundamental AffectNet-to-FER2013 shift.

Four — FLAN-T5 base, our generator, is intentionally minimal. As we saw on the replies slide, it drifts into multiple-choice or paraphrase formats on Sad and Surprise. A 1-billion-parameter instruction-tuned model is a one-line swap and would clearly help.

Five — and this is important — we didn't run a formal participant Likert study. The HCI component of this report is best read as expert review plus a ready-to-run study template that ships in the repository under *docs/HCI_STUDY_TEMPLATE.md* for the next group that picks this up.

TRANSITION

Let me close with the three numbers that tell most of the story.

19
of 20

Key Takeaways

Three numbers tell most of the story

ON SCREEN

Three large coloured stat cards: 23.1 → 68.5 percent face top-1 (visual fine-tune), 81.6 → 87.8 percent fused matched (now beats text alone), and 4.3 → 41.8 percent fused dissonant (face overrides wrong text). Below them, an italic synthesis sentence.

WHAT TO SAY

Three numbers. Just three.

On the left: 23.1 to 68.5 percent. That's what the visual fine-tune did to the face branch on its own. Plus 45 points, three-fold improvement, near-chance to usable.

In the middle: 81.6 to 87.8 percent. That's what the fine-tune did to the matched bimodal benchmark — fused now beats text alone, not just matches it.

On the right: 4.3 to 41.8 percent. That's what the fine-tune did to the dissonance benchmark — the fused decision can now override planted-wrong text on roughly 40 percent of cases instead of essentially never.

And the line below those three cards is the synthesis: the pipeline works, the trust knob is interpretable, and the visual fine-tune flipped the system from text-dominant into genuinely bimodal — at zero change to the deployed code path beyond a checkpoint swap.

TRANSITION

And with that, we're at the conclusion.

20
of 20

Thank you

Conclusion · Q&A · reproducibility pointers

ON SCREEN

Dark navy thank-you slide with a coral accent. "Questions, demo requests, and brutal critique welcome." Below: reproducibility pointers — the four scripts that produce every number in the deck. Below that: the datasets and models we used.

WHAT TO SAY

That's the whole story. We built the course's bimodal affective alignment stack — a transparent trust knob, a shared seven-class label space, late fusion, an instruction-tuned response generator, and a Streamlit interface that exposes every intermediate signal rather than hiding it.

Every single number in this deck is produced by a script in our repository. The four key ones are listed at the bottom: *run_evaluation* for the matched benchmark, *finetune_fer2013* for the visual fine-tune, *eval_dissonant_fer2013* for the dissonance probe, and *benchmark_latency* for the timings. They all read the same .env file and write both a JSON summary and a LaTeX fragment so the report rebuilds itself.

The datasets and models are public — GoEmotions, FER2013, RoBERTa-base, ResNet-50, and FLAN-T5 base. All pulled from HuggingFace on first use.

Thank you for listening. We'd love your questions, and if anyone wants a live demo of the Streamlit interface afterward — sliding alpha, watching the dissonance badge change, the whole thing — we have that ready.